

# LIMIT-SKETCHABLE INFINITY CATEGORIES

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# Presentability

A (locally small) category  $C$  is **accessible** if there exists a regular cardinal  $\kappa$  such that:

- ✓  $C$  admits  $\kappa$ -filtered colimits.
- ✓ There is a set of  $\kappa$ -compact objects that generate  $C$  under  $\kappa$ -filtered colimits.

A category is **(locally) presentable** if it is accessible and cocomplete.

| Presentable | Accessible not presentable | Not accessible |
|-------------|----------------------------|----------------|
|-------------|----------------------------|----------------|

|             |  |  |
|-------------|--|--|
| Set         |  |  |
| Grp         |  |  |
| sSet        |  |  |
| RMod        |  |  |
| Ch(RMod)    |  |  |
| Sheaf topos |  |  |

|                |  |
|----------------|--|
| Fields         |  |
| Hilbert spaces |  |

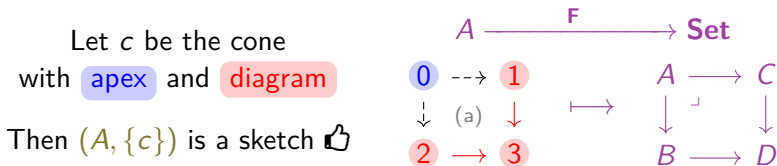
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| Top |
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# Limit sketches

A **limit sketch** (Bastiani and Ehresmann 1972) is a pair  $(A, C)$  of a small category  $A$  and a set of cones  $C$  over  $A$ .

A **model** of a limit sketch is a functor  $F : A \rightarrow \mathbf{Set}$  which sends cones of  $C$  to limits of  $\mathbf{Set}$ . A category is **limit-sketchable** if it is equivalent to the category of models of some limit sketch.

**Example.** Let  $A$  be the small category generated by the square (a).



A model  $F$  of the sketch  $(A, \{c\})$  is a pullback of sets 👍

# Representation theorem

## Theorem (Adamek and Rosicky 1994)

*The following are equivalent:*

- (i) *Presentable categories.*
- (ii) *Accessible localizations of presheaves over a small category.*
- (iii) *Limit-sketchable categories.*

## Theorem (Simpson 1999 and Lurie 2009)

*The following are equivalent:*

- (i) *Presentable  $\infty$ -categories.*
- (ii) *Accessible localizations of presheaves over a small  $\infty$ -category.*

### Goal

Presentable  $\infty$ -categories  $\stackrel{?}{\simeq}$  Limit-sketchable  $\infty$ -categories

# Plan

Presentable  $\infty$ -categories

Limit  $\infty$ -sketches

Representation theorem

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# Review of higher categories

Informally, an  $\infty$ -category (or  $(\infty, 1)$ -category) has objects and:

- $n$ -morphisms between  $(n - 1)$ -morphisms  $\forall n \geq 1$ ,
- $\circ$  and  $\text{id}$  are weak up to a higher morphism,
- $n$ -morphisms are invertible up to higher morphisms  $\forall n > 1$ .

In the Joyal-Lurie model of higher categories:

$\infty$ -**category** := Quasicategory ( $\mathbf{sSet}$  with weak composition)

$\infty$ -**groupoid** := Kan complex ( $\mathbf{qCat}$  with all morphisms invertible)

In a quasicategory the 0-simplices represent the objects, and the  $n$ -simplices represent  $n$ -morphisms.

## Example

- (a) Any 1-category can be viewed as an  $\infty$ -category using the nerve functor  $N : \mathbf{Cat} \rightarrow \mathbf{sSet}$ .
- (b) The  $\infty$ -category of homotopy types  $\mathcal{S}$ .

## Limits and colimits

Let  $\mathcal{C}$  be an  $\infty$ -category, and  $I$  be a small simplicial set. Given any object  $x \in \text{Obj}(\mathcal{C})$ , the **constant diagram**  $\Delta_x : I \rightarrow \mathcal{C}$  sends all objects of  $I$  to  $x$ , and all higher morphisms to higher identities.

Let  $D : I \rightarrow \mathcal{C}$  be a diagram and  $y \in \text{Obj}(\mathcal{C})$  be an object of  $\mathcal{C}$ . A natural transformation  $\alpha : \Delta_y \Rightarrow D$  **exhibits  $y$  as a limit of  $D$**  if, for all  $x \in \text{Obj}(\mathcal{C})$ ,  $\alpha$  induces an equivalence

$$\text{Map}_{\mathcal{C}}(x, y) \xrightarrow{\sim} \text{Cones}(I, D, x) := \text{Map}_{\text{Fun}(I, \mathcal{C})}(\Delta_x, D).$$

**Cocones** and **colimit cocones** are defined as cones and limit cones in the opposite  $\infty$ -category.



# Size of $\infty$ -categories

Let  $\kappa$  denote a regular cardinal and  $\mathcal{C}$  an  $\infty$ -category.

- A simplicial set is  $\kappa$ -**small** if it has less than  $\kappa$  non-degenerate simplices.
- An  $\infty$ -category is **small** if it is  $\kappa$ -small as a simplicial set for some regular cardinal  $\kappa$ .
- $\mathcal{C}$  is **essentially small** if there exists a categorical equivalence  $\mathcal{C}' \rightarrow \mathcal{C}$ , where  $\mathcal{C}'$  is a small  $\infty$ -category.
- $\mathcal{C}$  is **locally small** if for all objects  $x, y \in \text{Obj}(\mathcal{C})$  the mapping space  $\text{Map}_{\mathcal{C}}(x, y)$  is essentially small.

# Accessibility

- An  $\infty$ -category  $\mathcal{K}$  is  **$\kappa$ -filtered** if, for every  $\kappa$ -small simplicial set  $I$ , every diagram  $D : I \rightarrow \mathcal{K}$  admits a cocone  $\alpha : D \Rightarrow \Delta x$ .
- $\mathcal{C}$  admits  **$\kappa$ -filtered colimits** if it admits  $\mathcal{K}$ -indexed colimits, for every  $\kappa$ -filtered  $\infty$ -category  $\mathcal{K}$ .
- An object  $x \in \text{Obj}(\mathcal{C})$  is called  **$\kappa$ -compact** if the mapping space functor  $\text{Map}_{\mathcal{C}}(x, -) : \mathcal{C} \rightarrow \mathcal{S}$  preserves  $\kappa$ -filtered colimits. Denote by  $\mathcal{C}_{\kappa}$  the full sub- $\infty$ -category of  $\kappa$ -compact objects.

An  $\infty$ -category  $\mathcal{C}$  is **accessible** if it is locally small and there is a regular cardinal  $\kappa$  such that:

- ✓  $\mathcal{C}$  admits  $\kappa$ -filtered colimits.
- ✓  $\mathcal{C}_{\kappa}$  is essentially small and generates  $\mathcal{C}$  under  $\kappa$ -filtered colimits.

# Presentability

## Definition

An  $\infty$ -category is **presentable** if it is accessible and cocomplete.

## Example

- (a)  $\mathcal{S}$  is presentable.
- (b) Any  $\infty$ -topos is a presentable  $\infty$ -category.
- (c) Any presentable 1-category is also a presentable  $\infty$ -category.
- (d) If  $\mathcal{A}$  is a small  $\infty$ -category and  $\mathcal{C}$  is a presentable  $\infty$ -category, then  $\mathrm{Fun}(\mathcal{A}, \mathcal{C})$  is presentable.
- (e) In particular,  $\mathrm{PSh}(\mathcal{A}) := \mathrm{Fun}(\mathcal{A}^{\mathrm{op}}, \mathcal{S})$  is presentable.

# Characterization of presentability

A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a **localization** if it has a fully faithful right adjoint:

$$\mathcal{D} \begin{array}{c} \xleftarrow{F} \\ \perp \\ \xrightarrow{U} \end{array} \mathcal{C}$$

A localization is accessible if  $\mathcal{C}$  and  $\mathcal{D}$  are accessible.

If  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a localization, then  $\mathcal{D} \simeq \mathcal{L}_W \mathcal{C}$  where

- $\mathcal{L}_W \mathcal{C}$  is the  $\infty$ -category that **universally inverts**  $W$ .
- $W$  is the class of maps  $\eta_X : X \rightarrow UF(X)$ ,  $\forall X \in \mathcal{C}$ .

## Theorem (Simpson 1999 and Lurie 2009)

*The following are equivalent:*

- Presentable  $\infty$ -categories.*
- Accessible localizations of presheaves over a small  $\infty$ -category.*

# Plan

Presentable  $\infty$ -categories

Limit  $\infty$ -sketches

Representation theorem

## Limit $\infty$ -sketches

A **limit  $\infty$ -sketch** (Joyal 2008)  $\mathcal{T} = (\mathcal{K}, \mathfrak{L})$  is a small  $\infty$ -category  $\mathcal{K}$  equip with a set  $\mathfrak{L}$  of cones.

Let  $\mathcal{C}$  be a complete  $\infty$ -category. A functor  $F : \mathcal{K} \rightarrow \mathcal{C}$  is a **model** of a limit  $\infty$ -sketch  $\mathcal{T} = (\mathcal{K}, \mathfrak{L})$  in  $\mathcal{C}$  if it sends each cone in  $\mathfrak{L}$  to a limit cone in  $\mathcal{C}$ .

$\text{Mod}(\mathcal{T}, \mathcal{C}) := \infty\text{-category of models of } \mathcal{T} \text{ in } \mathcal{C}$

$\text{Mod}(\mathcal{T}) := \infty\text{-category of models of } \mathcal{T} \text{ in } \mathcal{S}$

We say that an  $\infty$ -category is **limit  $\infty$ -sketchable** (or **essentially  $\infty$ -algebraic**) if it is equivalent to the  $\infty$ -category of models of some limit  $\infty$ -sketch.

## Examples: Pointed objects

Let  $\mathcal{A}$  be the nerve of the small category generated by the graph

$$0 \xrightarrow{a} 1 ,$$

and  $\mathfrak{L}$  be the set of a cone with the empty diagram  $D : \emptyset \rightarrow \mathcal{A}$ , the apex  $0 \in \mathcal{A}$ , and the unique natural transformation  $\alpha : \Delta 0 \Rightarrow D$ .

Then  $\mathcal{T} = (\mathcal{A}, \mathfrak{L})$  is a limit  $\infty$ -sketch, and a model  $F : \mathcal{A} \rightarrow \mathcal{C}$  sends

$$(a : 0 \rightarrow 1) \longmapsto (F(a) : 1_{\mathcal{C}} \rightarrow F(1)),$$

where  $1_{\mathcal{C}}$  is the terminal object of  $\mathcal{C}$ .

Therefore,  $\text{Mod}(\mathcal{T}, \mathcal{C})$  is the  $\infty$ -category of **pointed objects** in  $\mathcal{C}$ .

## Examples: $\infty$ -algebraic theories

An  **$\infty$ -algebraic theory** (or  **$\infty$ -Lawvere theory**) is a small  $\infty$ -category with finite products. A **model** (or **algebra**) for an  $\infty$ -algebraic theory  $\mathcal{A}$  is a functor  $\mathcal{A} \rightarrow \mathcal{S}$  that preserves products.

Any  $\infty$ -algebraic theory is an  $\infty$ -sketch with only product cones

**Example.** Monoid objects ( $A_\infty$ -spaces), commutative monoid objects ( $E_\infty$ -spaces), group objects ( $\infty$ -groups),  $R$ -modules, ...

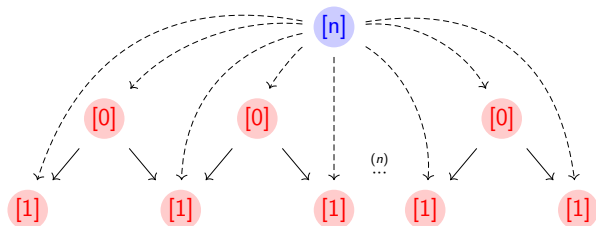
**Theorem** (Rosicky 2007 and Lurie 2009)

*The  $\infty$ -category of models of an  $\infty$ -algebraic theory is presentable.*



## Examples: Pre-category objects

Let  $\mathcal{C}$  be a complete  $\infty$ -category,  $\mathcal{A}$  be the nerve of  $\Delta^{\text{op}}$ , and  $c_n$  be the cone with **apex** and **diagram** for all  $n \in \mathbb{N}$ :



Then  $\mathcal{T} = (\mathcal{A}, \{c_n \mid n \in \mathbb{N}\})$  is a limit  $\infty$ -sketch, and a model  $F : \mathcal{A} \rightarrow \mathcal{C}$  is a simplicial object in  $\mathcal{C}$  such that

$$F_n \xrightarrow{\sim} F_1 \times_{F_0} F_1 \times_{F_0} \cdots \times_{F_0} F_1. \quad (\text{Segal condition})$$

Therefore,  $\text{Mod}(\mathcal{T}, \mathcal{C})$  is the  $\infty$ -category of **pre-category objects** in  $\mathcal{C}$ , and  $\text{Mod}(\mathcal{T})$  is the  $\infty$ -category of **Segal spaces**.

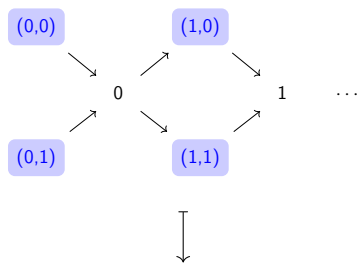
## Examples: Pre-spectrum objects

Let  $\mathcal{A}$  be the nerve of the small category generated by:

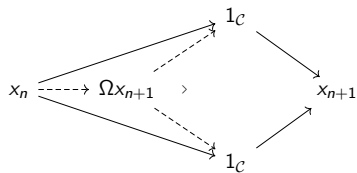
Let  $c_n$  be the cone with empty diagram and **apex**. Then

$$\mathcal{L} = \{c_n \mid n \in \mathbb{N}\}, \text{ and}$$

$$\mathcal{T} = (\mathcal{A}, \mathcal{L}) \text{ is an } \infty\text{-sketch.}$$



A model for  $\mathcal{T}$  is a sequence of pointed objects  $1 \rightarrow x_n$  for every  $n \in \mathbb{N}$ , and maps  $x_n \rightarrow \Omega x_{n+1}$ .



$\text{Mod}(\mathcal{T}, \mathcal{C}) \simeq$  **Pre-spectrum objects**

$\text{Mod}(\mathcal{T}) \simeq$  **Pre-spectra**

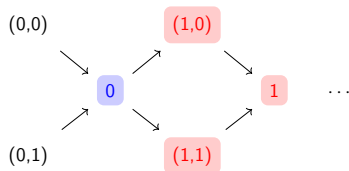
## Examples: Spectrum objects

Let  $\mathcal{A}$  be the nerve of the small category generated by:

Let  $d_n$  be the cone with  
diagram and apex. Then

$$\mathcal{L}' = \mathcal{L} \sqcup \{d_n \mid n \in \mathbb{N}\}, \text{ and}$$

$$\mathcal{T}' = (\mathcal{A}, \mathcal{L}') \text{ is an } \infty\text{-sketch.}$$

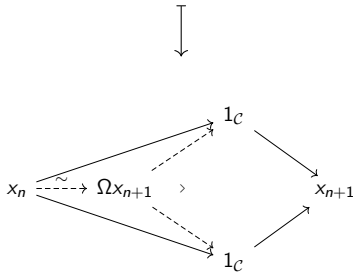


A model for  $\mathcal{T}'$  is a pre-spectrum object in  $\mathcal{C}$  such that

$$x_n \simeq \Omega x_{n+1}.$$

$$\text{Mod}(\mathcal{T}, \mathcal{C}) \simeq \mathbf{\text{Spectrum objects}}$$

$$\text{Mod}(\mathcal{T}) \simeq \mathbf{\text{Spectra}}$$



# Plan

Presentable  $\infty$ -categories

Limit  $\infty$ -sketches

Representation theorem

# Representation theorem

## Theorem (M.)

*An  $\infty$ -category is presentable  $\iff$  it is limit  $\infty$ -sketchable.*

## Corollary

*The models of a limit  $\infty$ -sketch in any presentable  $\infty$ -category are presentable.*

## Idea of the proof

- $\implies$  By Lurie, we know that a presentable  $\infty$ -category is equivalent to the  $\infty$ -category of Ind-objects for some small  $\infty$ -category. Then, use this characterization to construct a limit  $\infty$ -sketch.
- $\impliedby$  Prove that a model of any limit  $\infty$ -sketch is equivalent to an  $M$ -local object for a set of morphisms  $M$  in the presentable  $\infty$ -category  $\mathrm{Fun}(\mathcal{A}^{\mathrm{op}}, \mathcal{S})$ . Then, by Lurie, it is presentable.

## Future work

- 💡 Proof the **model categorical version** of the representation theorem, replacing presentable  $\infty$ -categories by combinatorial model categories.

Some work already done by Rosicky, but in different contexts!

- 💡 **Generalization:** A  $\infty$ -category is accessible if, and only if, it is equivalent to the  $\infty$ -category of models of an  $\infty$ -sketch.

A **sketch** is a limit sketch with a set of cocones which are sent to colimit cocones by any model.

- 💡 Accessibility, presentability, sketches, and representation theorem for  $\infty$ -**cosmoi** (Riehl-Verity 2022)

$\implies$  **Model-independent** version of this presentation!

# References

- Adamek, Jiří and Jiří Rosicky (1994). *Locally Presentable and Accessible Categories*. Vol. 189. Cambridge University Press.
- Simpson, Carlos (1999). “A Giraud-type characterization of the simplicial categories associated to closed model categories as  $\infty$ -pretopoi”. In: [arXiv: math/9903167](#).
- Lurie, Jacob (2009). *Higher Topos Theory*. *Annals of Mathematics Studies* 170. Princeton University Press.
- Joyal, André (2008). *The Theory of Quasi-Categories and its Applications*. *Barcelona: Lectures at CRM*.
- Rosicky, Jiří (Oct. 2007). “On homotopy varieties”. In: *Advances in Mathematics* 214.2, pp. 525–550.

**Thank you for listening!**

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